COURSE TITLE : METROLOGY LABORATORY
COURSE CODE : 3119
COURSE CATEGORY : B
SEMESTER : 3
PERIODS PER WEEK : 3
PERIODS PER SEMESTER : 45
CREDITS : 2

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Practicals in Metrology Lab (Linear measurements, Calibration)	20
2	Practicals in Metrology Lab (Angular Measurements, Pitch, Thread)	15
3	Study of gauges and comparators	10
	Total	**45**

COURSE OUTCOMES

After completion of the study the students will be able to:

1. Understand Linear measurement and Calibration
2. Practice on angular measurements,pitch, and thread.
3. Practice on gauges and comparators.

A. LIST OF PRACTICALS IN METROLOGY LABORATORY (At least 8 Experiments are to be done)

1. To measure the length, breath, thickness, depth and height with Vernier Caliper
2. To measure the length, breath, thickness, depth and height with Vernier Height Gauge
3. To measure the length, breath, thickness, depth and height with Micrometer
4. Calibration of Vernier Caliper
5. Caliberation of Vernier Heigt Gauge
6. Calibration of micrometer
7. Caliberation of Dial Gauge
8. Work Dialing with Dial Gauge
9. Measurement of angle with the help of Sine bar / Vernier Bevel protractor
10. To measure the diameter of a hole with the help of precision balls
11. To measure external and internal tapes with the help of taper gauges, Precision rollers and precision balls
12. To measure the pitch, angle and identify the form of thread of a screw
13. Check the accuracy of a Plug Gauge with Slip Gauge
14. Check the accuracy of a Plug Gauge with Micrometer
15. Study & sketch of various types of comparators and use them for comparing length of given pieces
16. Study and sketch of various types of optical projectors
17. Study of a Tool maker’s microscope